

Numerical Laboratory Activity 1: Continuous–Time Speed Control of a DC Motor

Exercise 1: *Ideal vs real derivative filter*

The derivative of a generic signal $u(t)$ can be obtained by applying an *ideal derivative* filter with transfer function $H(s) = s$. However, this filter is not physically implementable because its transfer function is not proper, meaning the system is not causal. To obtain a physically implementable version (*real derivative* filter), poles are introduced into the transfer function of the ideal filter until it becomes proper or strictly proper. These poles are added in a way that minimally alters the frequency response of the ideal filter, particularly within the frequency range where the signal spectrum is most significant. In typical control systems, signals are concentrated in the low-frequency range, so the poles must be placed at high frequencies to preserve the desired behavior.

Consider the following two *real derivative* filters:

$$H_1(s) = \frac{s}{Ts + 1}, \quad H_2(s) = \frac{\omega_c^2 s}{s^2 + 2\delta\omega_c s + \omega_c^2} \quad \text{con} \quad T = \frac{1}{2\pi 50}, \quad \omega_c = 2\pi 50, \quad \delta = \frac{1}{\sqrt{2}}$$

Using only the routines of the Control System Toolbox (CST), complete the following steps:

- 1) Plot the Bode diagrams of the two high-pass filters $H_1(s)$ e $H_2(s)$ on the same MATLAB figure.
- 2) Compare their frequency responses with that of the ideal derivative filter $H(s)$, to assess deviations in magnitude and phase.
- 3) Evaluate the responses of the two filters $H_1(s)$ e $H_2(s)$ to the sinusoidal input $u(t) = A \sin(\omega t)$, with $A = 1$ e $\omega = 2\pi 0.5$. Use equally spaced time instants over a 4 s time range, with a sampling interval of 1 ms. Display the two responses on a same MATLAB figure, and compare them with the ideal derivative of $u(t)$.
- 4) Repeat the previous step with $\omega = 2\pi 20$. Evaluate the amplification and phase lag introduced by the two filters at the frequency ω , and compare them with those of the ideal derivative filter. Determine which filter better approximates the ideal magnitude response and which provides a closer phase response.
- 5) Evaluate the the responses of the two filters $H_1(s)$ e $H_2(s)$ to the noisy ramp input $u(t) = r(t) + n(t)$, where $r(t) = 25t$ is a linear ramp signal, and $n(t)$ is a random noise with uniform distribution over the interval $[-a, a]$ with $a = 0.001$. Use the same time instants as in the previous steps (Note: for the generation of the signal $n(t)$, use the rand routine).

Plot the output responses of the two filters on a MATLAB figure, and compute the variance of the output noise after discarding the initial transient. Identify which filter produces the least noisy output and discuss the implications for signal differentiation in the presence of noise.

Exercise 2: Simulation of a continuous-time speed control system for a DC motor using the CST

Consider the continuous–time speed control system for a DC motor illustrated in Fig. 1, where

$$P'(s) = \frac{\Omega(s)}{U_a(s)} = \frac{k_m}{T_m s + 1}, \quad P(s) = \frac{\Theta(s)}{U_a(s)} = \frac{1}{s} P'(s), \quad k_m = 5.2, \quad T_m = 0.03$$

are the continuous–time transfer functions of the DC motor, from the armature voltage u_a [V] to the shaft speed ω [rad/s] and position ϑ [rad], $C(s)$ is the transfer function of the continuous–time controller, and $H(s)$ is the transfer function of a continuous–time “real” derivative filter used to get a speed estimate $\omega_m(t)$ from the position measurement $\vartheta_m(t)$. The inputs $\omega^*(t)$, $d(t)$ and $n(t)$ are respectively the speed reference, the load disturbance (e.g. a voltage equivalent of a disturbance torque applied to the motor), and the output noise of the position sensor.

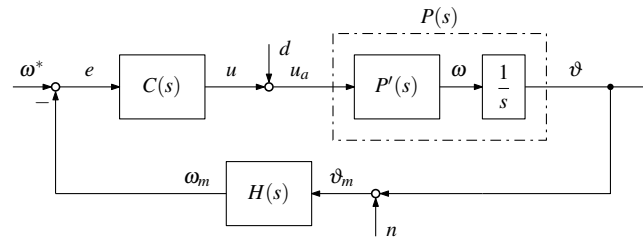


Figure 1: Continuous-time speed control system for a DC motor.

By using only the routines of the Control System Toolbox (CST), do the following steps:

- 1) Consider a purely proportional (P) controller with gain K_P , and $H_2(s)$ as the speed filter.

Plot the root locus of the control system with input $\omega^*(t)$ and output $\omega_m(t)$ as the proportional gain K_P varies from 0 to $+\infty$ (**Note**: when sketching the root locus, the *loop transfer function* must be considered). Then, identify the critical gain K_P^* beyond which the control system become unstable.

- 2) Set the proportional gain equal to $K_P = K_P^*/4$. Determine the actual speed $\omega(t)$, the measured speed $\omega_m(t)$, and the command $u(t)$ responses to a 300 rpm speed step reference, over a 0.3 s time range.

Show the results on a MATLAB figure, and determine the speed step response parameters (i.e. rise time, settling time at 5%, overshoot). Is the P controller capable of achieving zero steady-state tracking error (of the step reference input) ?

- 3) Consider now a continuous-time proportional-integral (PI) controller of the type:

$$C(s) = \frac{U(s)}{E(s)} = K_P \frac{s + 100}{s}$$

Repeat points 1) and 2). Determine whether the PI controller achieves zero steady-state tracking error.

Exercise 3: Simulation of a continuous-time speed control system for a DC motor using Simulink

Reconsider the continuous-time speed control system of exercise 2. By using Simulink, do the following points:

- 1) Implement a Simulink model of the speed control system, with the continuous-time P controller determined in point 2) of exercise 2.

Simulate the model with an appropriate solver (either variable-step, or fixed-step with a suitable step), by considering a step speed reference $\omega^*(t)$ of amplitude $A_\omega = 300$ rpm applied at $t_\omega = 0.1$ s, and a disturbance $d(t)$ of amplitude $A_d = -3$ V applied at $t_d = 0.5$ s.

On a MATLAB figure, plot the following signals:

- *plot 1*: actual speed $\omega(t)$, measured speed $\omega_m(t)$, speed reference $\omega^*(t)$, all in [rpm] units.
- *plot 2*: voltage command $u(t)$ [V].

Assess whether the P controller ensures perfect rejection of the constant disturbance $d(t)$.

- 2) Repeat the point 1), by using the PI controller obtained in point 3) of exercise 2.

Evaluate whether the PI controller ensures perfect rejection of the constant disturbance $d(t)$?